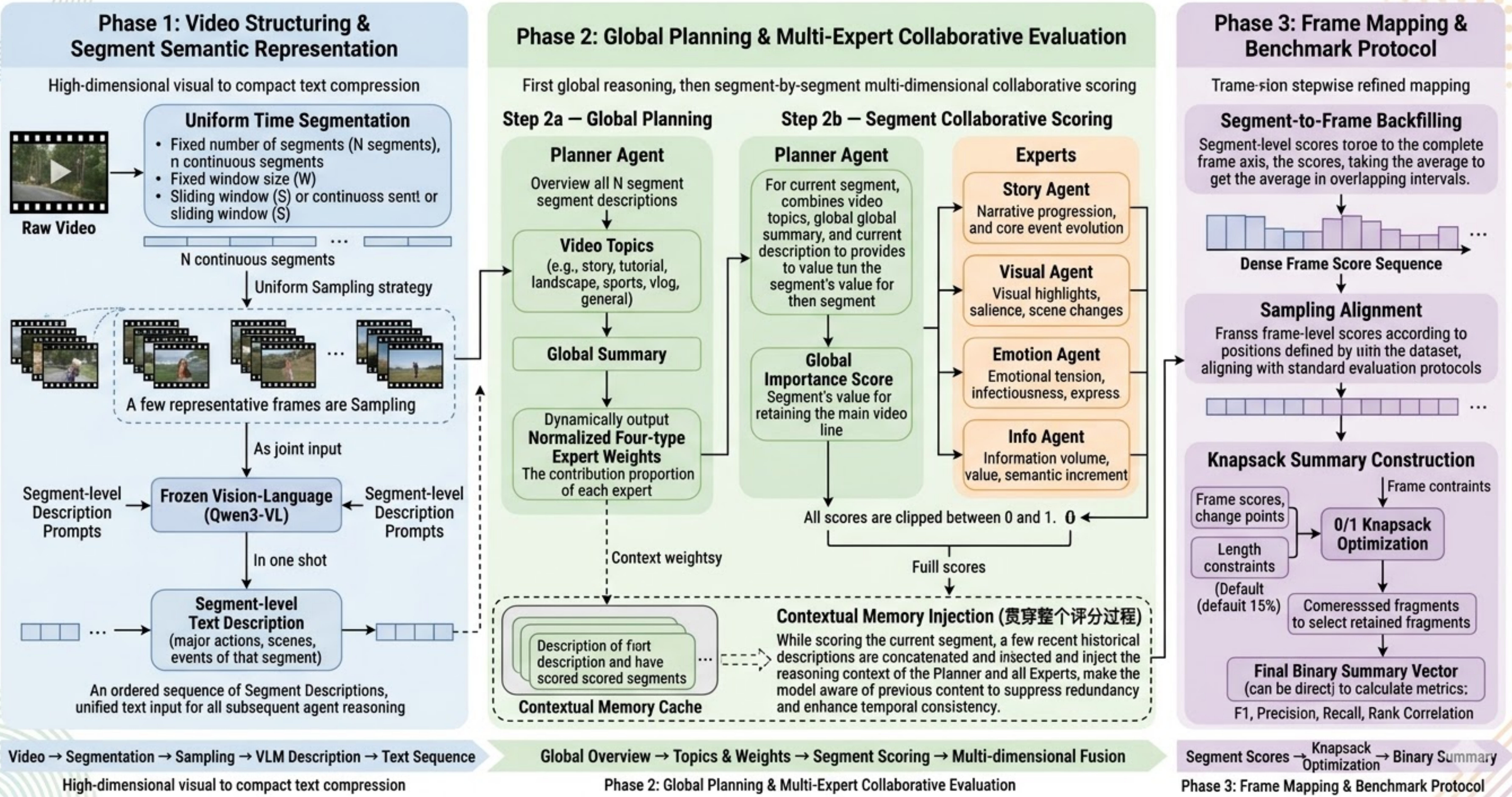
Phase 1: Video Structuring & Segment Semantic Representation
High-dimensional visual to compact text compression
Uniform Time Segmentation
• Fixed number of segments (N segments), n continuous segments
• Fixed window size (W)
• Sliding window (S) or continuoss sent! or sliding window (S)
Raw Video
N continuous segments
Uniform Sampling strategy
A few representative frames are Sampling
As joint input
Segment-level Description Prompts
Frozen Vision-Language (Qwen3-VL)
Segment-level Description Prompts
In one shot
Segment-level Text Description (major actions, scenes, events of that segment)
An ordered sequence of Segment Descriptions, unified text input for all subsequent agent reasoning
Phase 2: Global Planning & Multi-Expert Collaborative Evaluation
First global reasoning, then segment-by-segment multi-dimensional collaborative scoring
Step 2a — Global Planning
Planner Agent
Overview all N segment segment descriptions
Video Topics (e.g., story, tutorial, landscape, sports, vlog, general)
Global Summary
Dynamically output Normalized Four-type Expert Weights The contribution proportion of each expert
Context weightsy
Step 2b — Segment Collaborative Scoring
Planner Agent
For current segment, combines video topics, global global summary, and current description to provides to value tun the segment's value for then segment
Global Importance Score Segment's value for retaining the main video line
Experts
Story Agent
Narrative progression, and core event evolution
Visual Agent
Visual highlights, salience, scene changes
Emotion Agent
Emotional tension, infectiousness, express
Info Agent
Information volume, value, semantic increment
All scores are clipped between 0 and 1. 0
Fuill scores
Description of fiort description and have scored scored segments
Contextual Memory Cache
Contextual Memory Injection (贯穿整个评分过程)
While scoring the current segment, a few recent historical descriptions are concatenated and insected and inject the reasoning context of the Planner and all Experts, make the model aware of previous content to suppress redundancy and enhance temporal consistency.
Phase 3: Frame Mapping & Benchmark Protocol
Trame-sion stepwise refined mapping
Segment-to-Frame Backfilling
Segment-level scores toroe to the complete frame axis, the scores, taking the average to get the average in overlapping intervals.
Dense Frame Score Sequence
Sampling Alignment
Franss frame-level scores according to positions defined by uith the dataset, aligning with standard evaluation protocols
Knapsack Summary Construction
Frame contraints
Frame scores, change points
Length constraints
(Default (default 15%)
0/1 Knapsack Optimization
Comeresssed fragments to select retained fragments
Final Binary Summary Vector (can be directj to calculate metrics:
F1, Precision, Recall, Rank Correlation
Video → Segmentation → Sampling → VLM Description → Text Sequence
Global Overview → Topics & Weights → Segment Scoring → Multi-dimensional Fusion
Segment Scores → Knapsack Optimization → Binary Summary
High-dimensional visual to compact text compression
Phase 2: Global Planning & Multi-Expert Collaborative Evaluation
Phase 3: Frame Mapping & Benchmark Protocol